Unmixing-Based Spatiotemporal Image Fusion Accounting for Complex Land Cover Changes

Xiaolu Jiang and Bo Huang

***Abstract*— Spatiotemporal reflectance fusion has received considerable attention in recent decades. However, various challenges remain despite varying levels of success, especially regarding the recovery of spatial details with complex land cover changes. Taking the blending of Landsat and Moderate Resolution Imaging Spectroradiometer (MODIS) images as an example, this article presents a locally weighted unmixing-based spatiotemporal image fusion model (LWU-STFM) that focuses on recovering complex land cover changes. The core idea is to redefine the land use class of each pixel featuring land cover change at the prediction date. The spatial unmixing process is enhanced using a proposed geographically spectrum-weighted regression (GSWR), and then, we optimize similar neighboring pixels for the final weighted-based prediction. Experiments are conducted using semisimulated and actual time-series Landsat–MODIS datasets to demonstrate the performance of the proposed LWU-STFM compared with the classic spatial and temporal adaptive reflectance fusion model (STARFM), flexible spatiotemporal data fusion (FSDAF), two enhanced FSDAF models (SFSDAF and FSDAF 2.0), and a virtual image pair-based spatiotemporal fusion model for spatial weighting (VIPSTF-SW). The results reveal that the proposed LWU-STFM outperforms the other five models with the best quantitative accuracy. In terms of the relative dimensionless global error (ERGAS) index, the errors of Landsat-like images generated using LWU-STFM are 2.8%–63.4% lower than those of other models. From visual comparisons, LWU-STFM predictions illustrate encouraging improvements in recovering spatial details of pixels with complex land cover changes in heterogeneous landscapes and, thus, advancing applications of spatiotemporal image fusion for continuous and fine-scale land surface monitoring.**

***Index Terms*— Change detection, geographically spectrum-weighted regression (GSWR), land cover changes, locally spatial unmixing, spatiotemporal image fusion (STIF).**

I. Introduction

TEMPORALLY dense high-spatial-resolution satellite images are vital to provide data support for monitoring land surface dynamics and atmospheric environments, such as land cover mapping [1], change detection [2], crop growth monitoring [3], flooding disaster risk evaluation [4], and air temperature modeling [5]. Nevertheless, despite successive improvements in satellite technology, the acquisition of high spatiotemporal resolution images remains a major challenge [6], [7]. Therefore, current remotely sensed data from a single satellite sensor cannot meet the increasing demands of practical applications. In recent decades, spatiotemporal fusion of multisource satellite imagery has been largely developed to enhance the spatial and temporal resolutions of available datasets, including historical time-series images [8], [9].

The goal of spatiotemporal image fusion (STIF) is to generate alternative remote sensing imagery with both high spatial and high temporal resolutions [10]. Based on one or multiple pairs of high- and low-spatial-resolution images at t_0 (the prior date) and the low-spatial-resolution image at t_p (the prediction date), the fine image with high spatial resolution at t_p can be generated through the recovery of complex land surface changes during the observation period. Landsat and Moderate Resolution Imaging Spectroradiometer (MODIS) imagery are commonly adopted to test the validity and applicability of various STIF models. Landsat images have a higher spatial resolution (30 m) but lower temporal resolution (16 days), while MODIS sensor provides daily images with lower spatial resolutions (250 m, 500 m, and 1 km). The Landsat-like images generated by the blending of Landsat and MODIS data have been proven valuable in diverse applications, such as land cover classification [11], dynamic monitoring of lake wetlands [12], and crop progress mapping [13].

Traditional unmixing- and weighted-based STIF models are theoretically reliable and computationally efficient. Zhukov *et al.* [14] first presented an unmixing-based STIF model based on the linear spectral mixture theory [15], [16]. The land use classification of the high-spatial-resolution image at t_0 is conducted for the spatial unmixing process. However, when land cover changes occur during the observation period, the land use classification result at t_0 could not be applicable for the t_p image. Thus, unmixing-based models were developed to mainly enhance the capability of predicting land cover changes by introducing time-series high-spatial-resolution images [17]–[19]. The first weighted-based model, the spatial and temporal adaptive reflectance fusion model (STARFM), was proposed by Gao *et al.* [20]. STARFM performs well in capturing phenological changes while indicating evident limits in recovering land cover changes [21]. Then, an enhanced STARFM (ESTARFM) was developed to treat heterogeneous land surfaces by introducing multiple high-spatial-resolution images before and after t_p [22], [23]. However, time-series high-spatial-resolution images are usually unavailable in practical remotely sensed applications. Therefore, with only one high- and low-spatial-resolution

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image pair at t_0, the hybrid flexible spatiotemporal data fusion (FSDAF) model was presented [24]. FSDAF spatially interpolates the low-spatial-resolution image at t_p to capture the land cover change information in a separate spatial prediction step. Then, several enhanced FSDAF models were developed to improve the accuracy of fused images by dealing with missing intraclass problems [25], [26]. An enhanced FSDAF that incorporates subpixel class fraction change information (SFSDAF) indicated accurate prediction for land cover changes in heterogeneous landscapes [27]. FSDAF 2.0 was proposed to improve the performance of the FSDAF model in retrieving land cover changes and preserving spatial details [28]. Additionally, Zhao *et al.* [29] devised a robust adaptive spatial and temporal fusion model (RASTFM) that reorganizes complex land surface changes into nonshape and shape changes to handle them differently. Wang *et al.* [30] presented a linear regression-based virtual image pair (VIP) concept to perform a flexible spatiotemporal fusion for unmixing- and weighted-based models, thus forming two versions of a virtual image pair-based spatiotemporal fusion for spatial unmixing (VIPSTF-SU) and a virtual image pair-based spatiotemporal fusion for spatial weighting (VIPSTF-SW). The VIPSTF models can be conducted with one or multiple input high- and low-spatial-resolution image pairs.

To better address the issues about the recovery of complex land cover changes, this article presents a locally weighted unmixing-based spatiotemporal image fusion model (LWU-STFM) to blend the Landsat and MODIS surface reflectance with only one input image pair at t_0. Unlike the traditional spatial unmixing process used in the STIF field, we treat pixels with and without land cover changes separately. Specifically, class fractions generated from the Landsat classification map at t_0 are utilized to conduct the first unmixing process. The land surface reflectance of the pixels without land cover changes at t_p can be solved. Then, the real land use class of each changed pixel at t_p is recomposed using the exact class of the most similar unchanged pixel. The land surface reflectance of pixels with land cover changes at t_p can be further solved in the second unmixing process. Additionally, this study presents an enhanced geographically spectrum-weighted regression (GSWR) to improve the accuracy of the spatial unmixing process with the optimized land use class number and moving window size. Then, we define the more reliable similar neighboring pixels for the final prediction based on both the Landsat image at t_0 and the unmixed Landsat-like image at t_p, which contributes to excluding the unapplicable similar pixels with land cover changes. In this study, one semi-simulated Landsat–MODIS-like image pair and actual time-series satellite imagery are tested. The proposed LWU-STFM is compared with the classic STARFM, FSDAF, an enhanced FSDAF that incorporates sub-pixel class fraction change information (SFSDAF), FSDAF 2.0, and VIPSTF-SW models to demonstrate the performance of various STIF models.

II. Methodology

The proposed LWU-STFM is a hybrid model that combines unmixing- and weighted-based ideas. LWU-STFM aims to improve the capability of traditional STIF models in recovering spatial details of complex land cover changes in heterogenous landscapes. To highlight our contributions, we first give a brief introduction to basic unmixing-based STIF models and then emphasize innovations of the proposed LWU-STFM.

A. Basic Spatial Unmixing Process for the STIF

In general, the unmixing-based STIF models follow the basic theory that the land surface reflectance of each pixel is the linear spectral mixture of various land use classes. Various supervised or unsupervised land use classification methods are applied to classify the high-spatial-resolution image at t_0. Then, the fractional coverage of each land use class within a low-spatial-resolution pixel is defined by counting the number of high-spatial-resolution pixels belonging to a certain class $c (c = 1, \ldots, n)$

$$f_{ic} = N_{ic}/m \tag{1}$$

where n identifies the total number of land use classes, and m is the number of high-spatial-resolution pixels in a low-spatial-resolution pixel. N_{ic} is the number of high-spatial-resolution pixels belonging to class c within the ith low-spatial-resolution pixel. And f_{ic} is the solved fractional coverage of class c within the ith low-spatial-resolution pixel.

With the solved class fractions, the relationship of land surface reflectance between a certain low-spatial-resolution pixel and corresponding multiple high-spatial-resolution pixels can be mathematically represented as

$$Y_i = \sum_{c=1}^{n} (f_{ic} \times X_{ic}) + \varepsilon_i \tag{2}$$

where Y_i represents the land surface reflectance of the ith low-spatial-resolution pixel, and X_{ic} is the reflectance of land use belonging to class c within the ith low-spatial-resolution pixel. ε_i is the error term. Then, X_{ic} can be solved using k $(k > n)$ equations (2).

Traditional unmixing-based STIF models locally solve the differences of land surface reflectance between t_0 and t_p with a moving window based on the land use classification map of the high-spatial-resolution image at t_0 [17]. However, the class fractions generated from the land use classification map at t_0 will be invalid for the t_p image when land cover changes occur during the observation period. Then, the FSDAF model states that the reflectance can be solved by selecting k purest low-spatial-resolution pixels of each given class on a global scale, which contributes to reducing the effects of land cover changes [24]. In fact, FSDAF did not directly drive the existing limitation of unmixing-based STIF models on capturing land cover changes. The global unmixing process causes new problems of missing intraclass variety to be solved in varying local areas [25].

B. Proposed Spatial Unmixing Process Accounting for Land Cover Changes

To avoid the problems of missing intraclass variety in a global unmixing framework, this study locally conducts the spatial unmixing process using a moving window. In particular, the size of the moving window is a crucial parameter that affects both the efficiency of the unmixing process and the accuracy of the unmixing results by combining the adaptative number of land use classes [31]. Therefore, this study

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optimizes the combination of the land use class number and moving window size through a series of trial-and-error experiments, and detailed comparison results and relevant analyses can be found in Section V.

Additionally, referring to the geographically weighted regression model [32], this article develops an adaptive GSWR method to improve the spatial unmixing accuracy [33]. Specifically, we introduce a weight function that combines the geographic distance and spectral similarity. Theoretically, the land surface similarity of various neighboring pixels with the target pixel follows distance-decay principles [31]. Spectral differences contribute to estimating internal variances in diverse low-spatial-resolution pixels. Therefore, this study presents the decreasing weight function that combines geographic distances and spectral differences

$$W_j = \exp(1/\mathrm{dis}_j) \times \exp(1/\mathrm{spec}_j) \tag{3}$$

where dis_j and spec_j are the Euclidian distance and reflectance differences between the target low-spatial-resolution pixel and its jth neighboring pixel, respectively. In particular, dis_j and spec_j are first normalized to the same range to avoid different orders of magnitude. In this study, the bandwidth is not adopted when solving the system of multiple equations because the spatial unmixing process is always conducted in a moving window with a certain size.

In this study, k-means clustering analysis is adopted to perform land use classification of Landsat images at t_0 in a band-by-band form [34]. The first spatial unmixing process separately unmixes the original land surface reflectance of low-spatial-resolution images at t_0 and t_p instead of solving their differences. Then, land use classes of pixels with land cover changes are recomposed to perform the second unmixing process for the t_p image. Specifically, an adaptive unsupervised change detection using combined principal component analysis and the k-means clustering method is conducted to identify the pixels with land cover changes [35] and then provides a change mask at a high-spatial-resolution scale. As shown in Fig. 1, areas with diagonal up borders illustrate land cover changes, while others represent unchanged patches. For the target pixel (star in Fig. 1) with land cover change occurring from t_0 [Fig. 1(a)] to t_p [Fig. 1(b)], we search the most spectrally similar pixel without land cover change in the whole t_p image [circle in Fig. 1(b)]. Accordingly, we can find the exact land use class from the t_0 image [circle in Fig. 1(a)], which could be identified as the real land use class of the target pixel in the t_p image. Therefore, a real land use classification map is generated for the t_p image to conduct the second spatial unmixing process. Finally, the spatially unmixed images for both t_0 and t_p are achieved by combining the results generated from the first and second spatial unmixing processes.

C. Enhanced Neighboring Predictions

Spatially unmixed images usually have block problems due to spectral discontinuities crossing boundaries of various low-spatial-resolution pixels, and thus, removing block effects could be meaningful to achieve a more accurate prediction for the high-spatial-resolution image at t_p [24]. Referring to the FSDAF model, the weighted sum of similar neighboring pixels is introduced to perform the final prediction. However, similar neighboring pixels defined using the single high-spatial-resolution image at t_0 are limited to capturing land cover changes. To solve this problem, we introduce new constraints using the standard deviation index of a spatially unmixed t_p image. Therefore, the intersection of similar neighboring pixels identified from the original high-spatial-resolution image at t_0 and spatially unmixed t_p image is defined as the optimal similar neighboring pixels. Finally, no more than 30 spectrally similar neighboring pixels are used for the neighboring prediction of each target pixel in this study. Additionally, benefiting from the conversion coefficient concept proposed in ESTARFM [22], this study takes spatially unmixed t_0 and t_p images as the corresponding high-spatial-resolution image pair to calculate various weights of similar neighboring pixels and conduct the linear regression, thus following the final prediction of the high-spatial-resolution image at t_p.

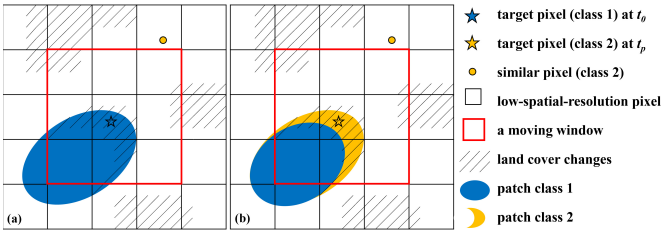


Fig. 1. Framework of recomposing land use class of the target pixel featuring land cover change. Images with a land cover change mask (a) t_0 and (b) t_p.

III. Data Description and Experimental Materials

This article indicates the performance of the proposed LWU-STFM using semisimulated and actual time-series Landsat–MODIS datasets, and the land surface reflectance in near-infrared (NIR), blue, and green bands is tested. All the actual Landsat images cover the size of 1200 pixels × 1200 pixels with a spatial resolution of 30 m, while both the simulated MODIS-like and actual satellite images captured from the MODIS sensor are resampled to a spatial resolution of 480 m using simple bilinear interpolation.

A. Semisimulated Landsat–MODIS-Like Image Pair

Complex factors related to the satellite technique, e.g., orbit parameters and imaging times, usually interfere with the spatiotemporal fusion of actual satellite images. Therefore, the simulated data have been popular to theoretically identify the capability of the proposed STIF models by eliminating environmental interference [36]. In this study, actual Landsat images are utilized as high-spatial-resolution images, while the corresponding MODIS-like images are generated from actual Landsat images. Accordingly, the semisimulated Landsat–MODIS-like image pairs are shown in Fig. 2.

The Landsat images are extracted from actual satellite data provided by Emelyanova *et al.* [37], which is located in the Coleambally Irrigation Area (CIA), a highly heterogeneous irrigated agricultural landscape, in southern New South Wales, Australia (34.0034°E, 145.0675°S), which grows various summer crops, such as rice, maize, sorghum, and soybean. The

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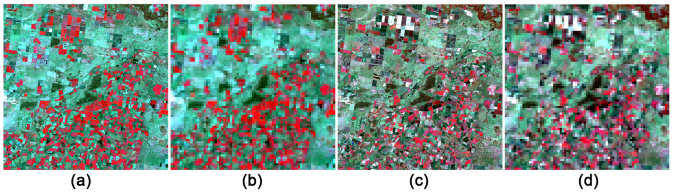


Fig. 2. Semisimulated Landsat–MODIS-like image pair in the CIA study site: including actual Landsat images captured on: (a) March 9, 2002 and (c) April 26, 2002 and the (b) and (d) corresponding downsampled MODIS-like images. All images are presented as Landsat bands 4, 1, and 2 displayed as RGB.

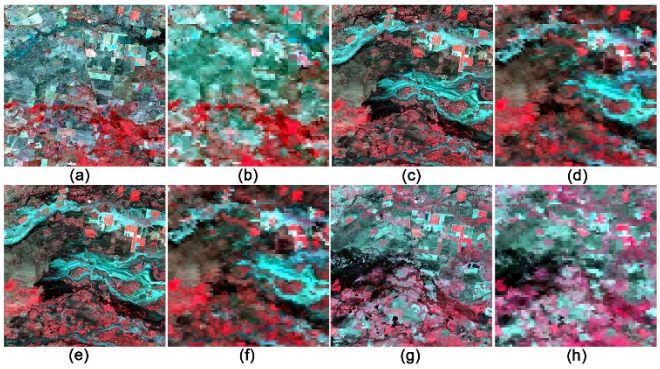


Fig. 3. Two cases of STIF using actual time-series Landsat–MODIS image pairs in the LGC study site. Case 1: actual Landsat images captured on: (a) November 26, 2004 and (c) December 12, 2004 and the (b) and (d) corresponding downsampled MODIS images. Case 2: actual Landsat images captured on: (e) December 12, 2004 and (g) December 28, 2004 and the (f) and (h) corresponding downsampled MODIS images. All images are presented as Landsat bands 4, 1, and 2 displayed as RGB.

reaping activities of grown crops occurred from March to the end of April 2002, and thus, a small parcel of croplands changed from vegetation to nonvegetation [29]. As shown in Fig. 2, the landscapes displayed on April 26, 2002 significantly differ from the images captured on March 9, 2002. The experiment to predict the Landsat image on April 26, 2002 is performed using the Landsat–MODIS-like image pair on March 9, 2002 and the MODIS-like image on April 26, 2002.

B. Actual Time-Series Satellite Images

To verify the performance of the proposed LWU-STFM on practical applications facing with complex land cover changes, actual time-series satellite images obtained from Landsat and MODIS sensors are experimented. Emelyanova *et al.* [38] provides 14 pairs of Landsat–MODIS imagery collected from the lower Gwydir catchment (LGC, 149.2815 E, 29.0855 S) study site in northern New South Wales, thus covering both summer and winter crop growing seasons from April 2004 to April 2005 [38]. An abrupt flooding disturbance caused large land cover changes in the mid-December 2004, and thus, the land surface reflectance dramatically changed before and after the flooding event. As shown in Fig. 3, three time-series Landsat–MODIS image pairs captured on November 26, 2004, December 12, 2004, and December 28, 2004 are adopted to form two cases of STIF.

C. Comparison and Quantitative Evaluation Indices

The proposed LWU-STFM essentially aims to overcome the limitations of traditional STIF models on the recovery of land cover changes using only one prior high- and low-spatial-resolution image pair. Therefore, this article designs comparison experiments with five advanced STIF models, including classic STARFM, FSDAF, SFSDAF, FSDAF 2.0, and VIPSTF-SW.

This study qualitatively and quantitatively compares the performance of the proposed LWU-STFM with those of five other models. With both the global view and highlighted local windows of fused images, a visual comparison is implemented with original satellite images [29]. Simultaneously, four indices for indicating spectral and spatial similarities between fused and actual Landsat images are used to perform the quantitative evaluation of various STIF models [39], including metrics of correlation coefficient (CC) and root-mean-square error (RMSE) featuring the spectral similarity, the structural similarity (SSIM) index [40] to indicate local patterns of pixel intensities in the spatial domain, and the relative dimensionless global error (ERGAS) [41] to reflect the quality situation of fused images on the whole in terms of the normalized average error.

IV. Experimental Results and Analyses

A. Image Fusion With One Semisimulated Image Pair

The quantitative evaluation shown in the first line in Table I indicates that the two original Landsat images captured on March 9, 2002 and April 26, 2002 in the CIA study site are very poorly correlated. The low relativity is caused by the reaping activities of grown crops at the end of April, which largely changed the land surface reflectance and led to a more heterogeneous landscape. Therefore, the image fusion task is relatively difficult in this case study. The quantitative evaluation results of Landsat-like images generated by various STIF models in Table I indicate that the proposed LWU-STFM outperforms the other five models with the highest CC and SSIM and lowest RMSE and ERGAS. For all three bands, LWU-STFM predictions are slightly better than the three FSDAF-based models (FSDAF, SFSDAF, and FSDAF 2.0) but far superior to STARFM and VIPSTF-SW. In terms of the ERGAS index, LWU-STFM globally improves the accuracy of fused images compared to other models by 3.7%–58.5%.

The Landsat-like images on April 26, 2002 generated by various STIF models are displayed for comparison with actual Landsat images on March 9, 2002 and April 26, 2002 in Fig. 4. From the visual comparison, LWU-STFM shows the most accurate Landsat-like image [Fig. 4(h)]. In particular, LWU-STFM accurately captures land cover changes and well recovers their spatial details. To highlight the performance of various STIF models in predicting land cover changes, zoomed-in view images of the subarea marked in Fig. 4 are presented in Fig. 5. From the original Landsat images captured on March 9, 2002 [Fig. 5(a)] and April 26, 2002 [Fig. 5(b)], a small parcel of land cover changes from cropland to bare land can be found in the subarea. The comparison results clearly indicate that STARFM, three FSDAF-based, and VIPSTF-SW models [Fig. 5(c)–(g)] cannot capture these small land cover changes. They generate Landsat-like images that are more similar to the prior Landsat image

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TABLE I
QUANTITATIVE EVALUATION WITH ONE SEMISIMULATED LANDSAT–MODIS-LIKE IMAGE PAIR IN THE CIA STUDY SITE

Method	CC			RMSE			SSIM			ERGAS
	NIR	Blue	Green	NIR	Blue	Green	NIR	Blue	Green	
HR_0	0.483	0.465	0.451	817.445	216.063	276.229	0.466	0.449	0.447	620.067
STARFM	0.729	0.578	0.734	455.632	163.590	163.917	0.728	0.574	0.725	413.294
FSDAF	0.840	0.803	0.833	345.990	99.036	132.001	0.821	0.794	0.824	281.865
SFSDAF	0.859	0.824	0.851	327.325	94.193	124.890	0.842	0.816	0.843	267.343
FSDAF 2.0	0.849	0.818	0.845	336.895	95.756	127.504	0.842	0.811	0.839	272.902
VIPSTF-SW	0.683	0.671	0.692	478.780	127.342	176.974	0.675	0.666	0.686	373.548
LWU-STFM	**0.861**	**0.841**	**0.868**	**325.801**	**90.112**	**118.258**	**0.858**	**0.836**	**0.864**	**257.435**

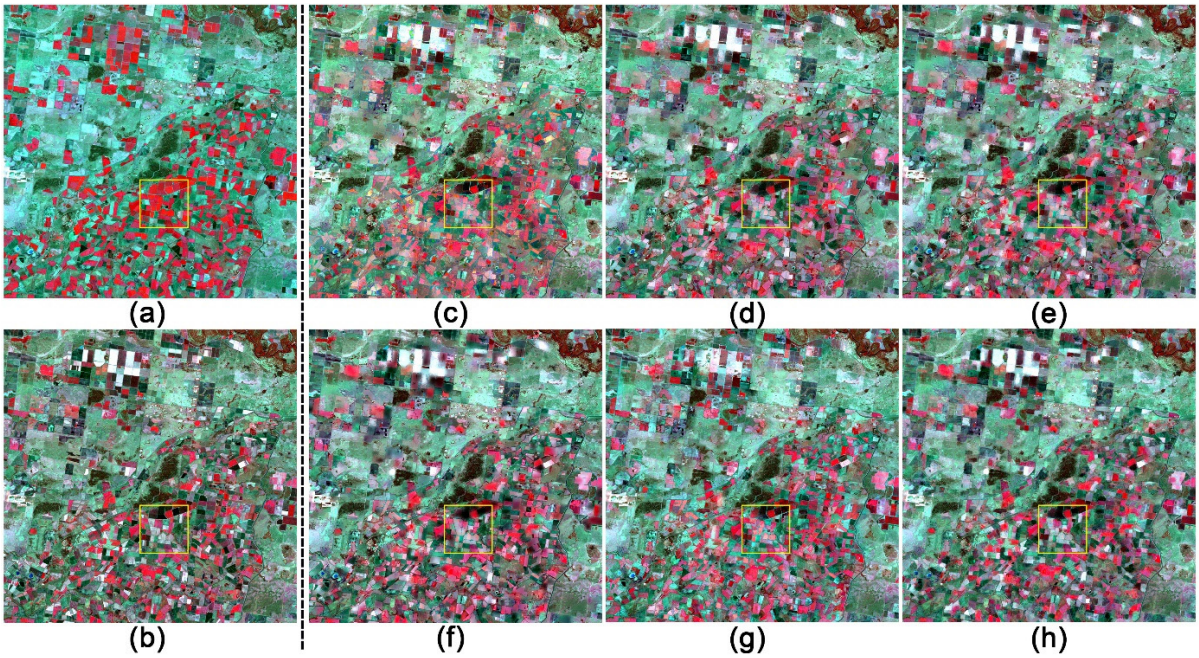


Fig. 4. Actual Landsat images in the CIA study site captured on: (a) March 9, 2002 and (b) April 26, 2002, and Landsat-like images on April 26, 2002 generated using various STIF models, including: (c) STARFM; (d) FSDAF; (e) SFSDAF; (f) FSDAF 2.0; (g) VIPSTF-SW; and (h) LWU-STFM. All images are presented as Landsat bands 4, 1, and 2 displayed as RGB.

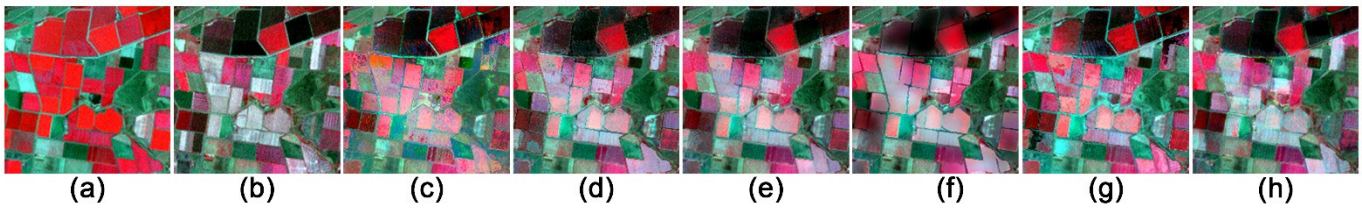


Fig. 5. Zoomed-in view images of the subarea marked in Fig. 4 extracted from actual Landsat images captured on: (a) March 9, 2002 and (b) April 26, 2002, and Landsat-like images on April 26, 2002 generated using various STIF models, including: (c) STARFM; (d) FSDAF; (e) SFSDAF; (f) FSDAF 2.0; (g) VIPSTF-SW; and (h) LWU-STFM. All images are presented as Landsat bands 4, 1, and 2 displayed as RGB.

on March 9, 2002 [Fig. 5(a)]. In contrast, the proposed LWU-STFM generates the best Landsat-like image [Fig. 5(h)] that successfully recovers the spatial details of land cover changes and well preserves clear boundaries of small land parcels.

B. Image Fusion With Actual Time-Series Satellite Datasets

Figs. 6 and 7 display the Landsat-like images in the LGC study site generated by various STIF models on December 12, 2004 and December 28, 2004, respectively. An abrupt flooding event in the mid-December 2004 led to complex land cover changes in the actual time-series satellite images. Thus, the identification of regions inundated by floods is a key criterion to verify the performance of varying STIF models. From the subarea A marked in Fig. 6, we can see that LWU-STFM indicates the proximal boundary for the inundated area [Fig. 6(h)], and following FSDAF and VIPSTF-SW predictions well preserve the spatial details [Fig. 6(d) and (g)]. By comparison, the STARFM, SFSDAF, and FSDAF 2.0 models produce blurrier images [Fig. 6(c), (e), and (f)]. In addition, land surface changes in the original Landsat images in subarea B [Fig. 6(a) and (b)] indicate complex flooding effects in neighbors. Here, the STARFM and VIPSTF-SW models seldom capture land cover changes [Fig. 6(c) and (g)]. LWU-STFM predictions

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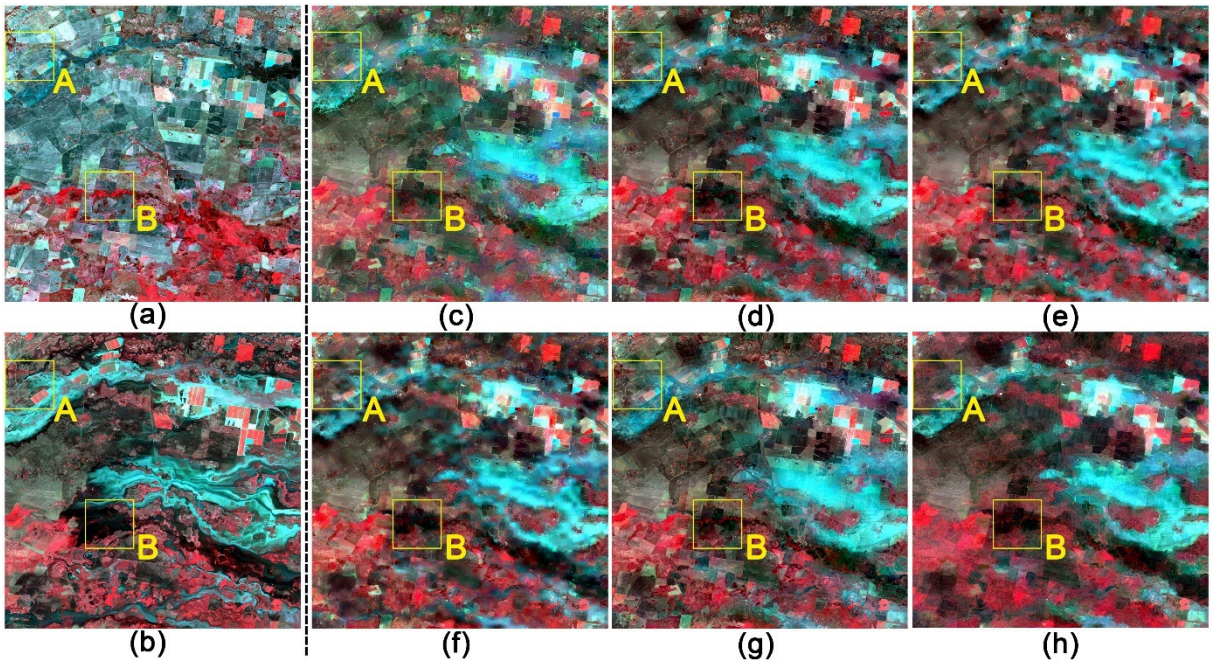


Fig. 6. Actual Landsat images in the LGC study site captured on: (a) November 26, 2004 and (b) December 12, 2004, and Landsat-like images on December 12, 2004 generated using various STIF models, including: (c) STARFM; (d) FSDAF; (e) SFSDAF; (f) FSDAF 2.0; (g) VIPSTF-SW; and (h) LWU-STFM. All images are presented as Landsat bands 4, 1, and 2 displayed as RGB.

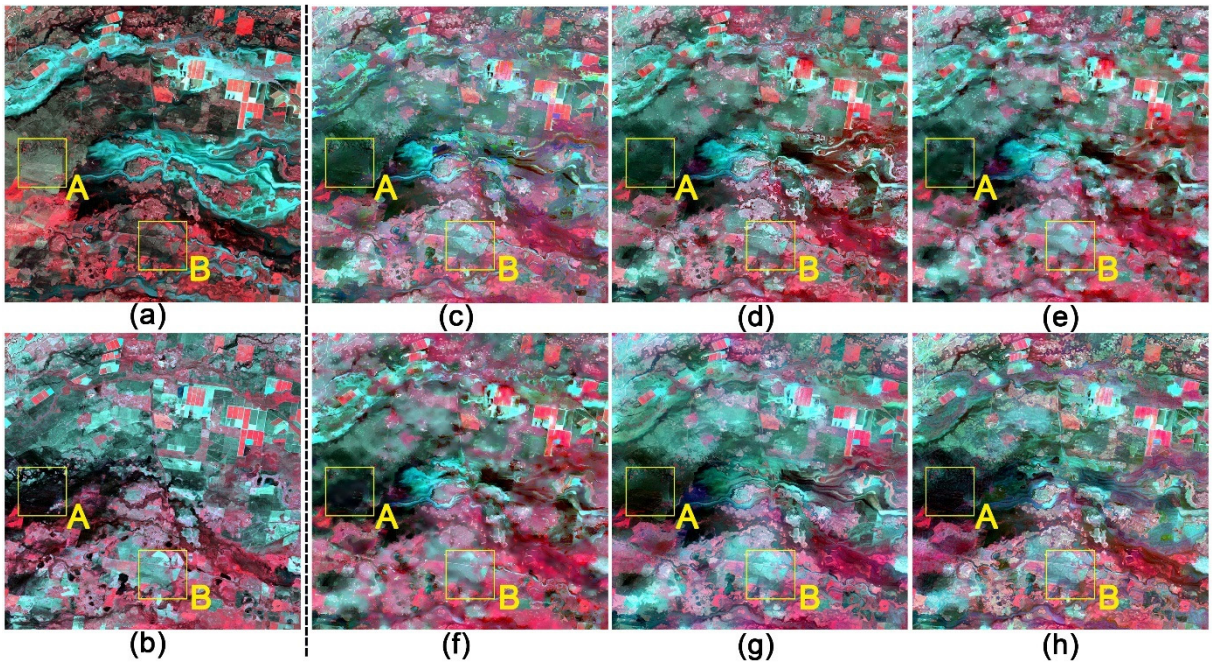


Fig. 7. Actual Landsat images in the LGC study site captured on: (a) December 12, 2004 and (b) December 28, 2004, and Landsat-like images on December 28, 2004 generated using various STIF models, including: (c) STARFM; (d) FSDAF; (e) SFSDAF; (f) FSDAF 2.0; (g) VIPSTF-SW; and (h) LWU-STFM. All images are presented as Landsat bands 4, 1, and 2 displayed as RGB.

[Fig. 6(h)] are slightly better than the three FSDAF-based models [Fig. 6(d)–(f)].

After the flooding event, the recovery of the land surface caused land cover changes to a certain degree, which can be found in the subarea A marked in Fig. 7. Obviously, the STARFM, FSDAF, and SFSDAF models generate images with visually large errors [Fig. 7(c)–(e)]. The FSDAF 2.0 and VIPSTF-SW predictions are relatively better [Fig. 7(f) and (g)], while compared with the actual Landsat image captured on December 12, 2004 [Fig. 7(a)], the closest Landsat-like image is still generated by the proposed LWU-STFM [Fig. 7(h)]. Additionally, subarea B in Fig. 7 is highlighted to show the performance of various STIF models in predicting the phenological changes that are not affected by the flooding event [Fig. 7(a) and (b)]. By visual comparison, the Landsat-like image generated using LWU-STFM illustrates the best spatial details with satisfactory spectral similarity [Fig. 7(h)], and following the STARFM prediction is basically accurate. The FSDAF and VIPSTF-SW models generate images with fewer spatial details [Fig. 7(d) and (g)],

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TABLE II
QUANTITATIVE EVALUATION WITH ACTUAL TIME-SERIES SATELLITE IMAGES IN THE LGC STUDY SITE

Case	Method	CC			RMSE			SSIM			ERGAS
		NIR	Blue	Green	NIR	Blue	Green	NIR	Blue	Green	
Case 1	STARFM	0.610	0.601	0.629	575.470	185.579	256.052	0.607	0.600	0.627	616.070
	FSDAF	0.764	0.642	0.651	436.294	176.834	248.884	0.739	0.639	0.649	543.996
	SFSDAF	0.757	0.639	0.655	441.780	179.588	249.226	0.733	0.636	0.653	549.398
	FSDAF 2.0	0.762	0.658	0.663	435.043	170.936	245.292	0.738	0.657	0.661	534.412
	VIPSTF-SW	0.734	0.655	0.671	465.871	166.795	235.024	0.719	0.648	0.660	534.967
	LWU-STFM	**0.764**	**0.677**	**0.674**	**439.442**	**164.196**	**232.431**	**0.739**	**0.670**	**0.661**	**519.553**
Case 2	STARFM	0.750	0.282	0.293	601.016	372.029	586.006	0.737	0.216	0.197	1123.231
	FSDAF	0.853	0.565	0.580	429.238	180.782	245.113	0.850	0.559	0.571	532.835
	SFSDAF	0.848	0.588	0.595	418.429	171.592	234.429	0.848	0.584	0.589	509.422
	FSDAF 2.0	0.846	0.574	0.576	410.839	177.319	230.737	0.846	0.567	0.573	512.623
	VIPSTF-SW	0.849	0.711	0.734	475.588	153.078	167.265	0.835	0.659	0.700	446.214
	LWU-STFM	**0.867**	**0.738**	**0.726**	**399.148**	**138.393**	**172.590**	**0.866**	**0.719**	**0.718**	**411.184**

while SFSDAF and FSDAF 2.0 illustrate overly smooth results [Fig. 7(e) and (f)]. Therefore, the visual comparison results in Figs. 6 and 7 generally conclude that LWU-STFM generates the best Landsat-like images in the two STIF cases in the LGC study site.

The quantitative evaluation results of Landsat-like images generated using various STIF models for actual time-series satellite images in the LGC study site are listed in Table II. Compared with the experimental results of the semisimulated Landsat–MODIS-like image pair in the CIA study site (Table I), here, the general accuracy is relatively low. Systemic differences in various satellite sensors lead to the spectral distortion between actual Landsat and MODIS data. Simultaneously, the larger and more complex land cover changes are caused by the abrupt flooding event.

For the two STIF cases in the LGC study site, the quantitative evaluation results of various STIF models vary. In case 1, LWU-STFM shows the highest CC and SSIM for all three bands. Although the RMSE is slightly higher than that of FSDAF for the NIR band, the global error of LWU-STFM prediction is still the lowest with the lowest ERGAS and average RMSE of the three spectral bands. Compared with other models, the ERGAS accuracy of LWU-STFM improves by 2.8%−15.7% for the prediction of the Landsat image on December 12, 2004.

For the prediction of the Landsat image on December 28, 2004 in case 2, LWU-STFM indicates the highest SSIM and the lowest ERGAS for all three bands, thus suggesting that the LWU-STFM prediction preserves the best spatial details and reduces the global error of Landsat-like images generated by other models by 7.9%–63.4%. For CC and RMSE, LWU-STFM generates the most accurate results in the NIR and blue bands, while the VIRSTF-SW prediction is slightly better in the green band. In terms of the average accuracy of all three bands, LWU-STFM generally outperforms the other models with the highest CC and lowest RMSE.

Given that the ERGAS is recognized as an integrated index to evaluate the whole quality of Landsat-like images generated by various STIF models, we can generally conclude that the proposed LWU-STFM has the best quantitative accuracy compared to the other five models for the prediction of Landsat images captured on December 12, 2004 and December 28, 2004.

V. DISCUSSION

In recent decades, abundant STIF models have been developed to blend the satellite imagery with various spatial and temporal resolutions, and these fused images from multisource satellite sensors could effectively increase the capability of current remote sensing images for long-term and fine-scale monitoring of land surface dynamics. In fact, common land surface conversions are complex in practical applications of remote sensing data. Natural phenological changes in forests or farmlands and complex land cover changes caused by urban expansion or abrupt disturbances usually exist in one certain landscape. Surely, a treatment to capture various land surface changes in a unified framework is crucial. However, existing STIF models still have large limitations on the effective recovery of spatial details with complex land cover changes. Therefore, this article proposes the LWU-STFM, which focuses on recovering complex land cover changes to unitedly address diverse land surface conversions with only one prior image pair. To clarify the procedure of the proposed model, here, we discuss several key parameters in the spatial unmixing process by showing some essential intermediate results.

A. Optimization of the Combination of Land Use Class Number and Moving Window Size in the Spatial Unmixing Process

The performance of unmixing-based STIF models largely depends on a fine land use classification map, which means that the optimized number of land use classes for a certain unsupervised class selector is crucial [31]. Therefore, the appropriate land use class number should be identified, which has been stated as 3, 4, 5, or 6 classes [24]. Additionally, the spatial unmixing process is locally conducted in a certain moving window, and thus, the size of the moving window is another key parameter for optimizing the unmixing process. In previous studies, the popular moving window size was

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TABLE III
SPATIALLY UNMIXED ACCURACY OF t_0 AND t_p IMAGES WITH VARYING LAND USE CLASS NUMBERS AND MOVING WINDOW SIZES WITH AVERAGE CC INDEX

Class number	Window size for unmixing t_0 image					Class number	Window size for unmixing t_p image				
	3 × 3	5 × 5	7 × 7	9 × 9	11 × 11		3 × 3	5 × 5	7 × 7	9 × 9	11 × 11
3	0.914	0.923	0.921	0.919	0.916	3	0.720	0.769	0.779	0.780	0.781
4	0.913	0.939	0.939	0.936	0.938	4	0.625	0.737	0.763	0.765	0.773
5	0.832	0.955	0.956	0.955	0.955	5	0.533	0.719	0.748	0.767	0.775
6	0.883	0.963	0.964	0.965	0.965	6	0.346	0.687	0.737	0.764	0.769

5 × 5 low-spatial-resolution pixels and few experiments have identified their effects on unmixing results [31]. In this study, considering the time cost, we design various sizes of the moving window to identify the effects for spatial unmixing accuracy, including 3 × 3, 5 × 5, 7 × 7, 9 × 9, and 11 × 11. With varying combinations of land use class number and moving window size, the spatial unmixing is conducted for both the t_0 and t_p images based on the land use classification map generated from the Landsat image at t_0. The accuracy of spatially unmixed images is indicated using the average CC index of all three spectral bands, which is shown in Table III.

With changes in land use class number and moving window size, the accuracy of spatially unmixed t_0 and t_p images shows varying trends. Specifically, as the land use class number increases, the accuracy of unmixed t_0 images is improved, while for t_p images, the accuracy generally decreases. In fact, when the land use classification result is sufficiently accurate, more classes contribute to enhancing the spatial unmixing process, which can interpret the trend indicated by the unmixing of the t_0 image in Table III. However, for the specific STIF applications, the low-spatial-resolution image at t_p is spatially unmixed using the land use classification result at t_0, which is not totally equal to the real land use classification at t_p even for the case without land cover changes. When land cover changes occur during the observation period, more land use classes will lead to more uncertainty. As shown in Table III, the accuracy of spatially unmixed t_p images becomes lower with more land use classes. Therefore, this study set the number of land use classes for local spatial unmixing as 3.

On the other hand, the size of the moving window slightly affects the spatial unmixing accuracy. As shown in Table III, the small moving window leads to very low accuracy due to insufficient equations. As the moving window size increases, the accuracy increases for both spatially unmixed t_0 and t_p images and gradually remains stable. Simultaneously, considering the more expensive time cost as the moving window size becomes larger, we set the moving window with a size of 11 × 11 low-spatial-resolution pixels. Therefore, the spatial unmixing process is performed with the optimized combination of three land use classes and the moving window size of 11 × 11 MODIS pixels in this study.

B. Procedure of Spatial Unmixing for Land Cover Changes

As shown in Fig. 8, referring to the original Landsat image captured on April 26, 2002 in the CIA study site, the intermediate results are presented to illustrate the specific procedure of spatial unmixing for land cover changes. With the optimized spatial unmixing process, including the appropriate land use classification selector, the optimized combination of land use class number and moving window size, and the enhanced GSWR method, the spatially unmixed Landsat-like image in the first unmixing process is displayed in Fig. 8(b). From the highlighted yellow circle, the problem that land cover changes cannot be captured can be easily found. Therefore, this study develops the second unmixing process to directly treat land cover changes.

Here, the specific steps to produce the land cover change mask [Fig. 8(c)] are clarified and analyzed. First, we conduct land cover change detection based on the low-spatial-resolution MODIS image pair instead of using the known high-spatial-resolution Landsat image at t_0. The land cover change detection technique is limited to obtain exact land cover change areas even with actual Landsat image pairs at t_0 and t_p. When we conduct the land cover change detection with MODIS images, the requirements for the capability of change detection will not be very strict. Additionally, the detected areas with land cover changes will become discrete if we use the known Landsat image at t_0 and then introduce unexpected noise in the subsequent land use class decomposition. On the other hand, we need the land cover change mask at a high-spatial-resolution scale to remain consistent with the high-spatial-resolution land use classification map at t_0. Therefore, the land cover change detection is designed using the resampled MODIS image pair using the bilinear interpolation method. Accordingly, the land cover change mask is generated, as shown in Fig. 8(c). With the land cover change mask, the actual land use class of each pixel featuring land cover change can then be recomposed using the exact class of the most spectrally similar pixel as the framework described in Fig. 1. Finally, the second unmixing process can be performed to update the land surface reflectance of the changed pixels, and the final unmixed image is shown in Fig. 8(d). From the visual comparison in the yellow circle, the second unmixing process based on the land cover change detection largely improves the capability of the spatial unmixing method to deal with the land cover changes.

Theoretically, the proposed LWU-STFM is a hybrid model that possesses several strengths of previous unmixing- and weighted-based STIF models. Simultaneously, the LWU-STFM aims to break through the limitations of traditional models to treat complex land cover changes better. Consequently, LWU-STFM achieves a higher accuracy and indicates encouraging improvements in recovering the spatial details

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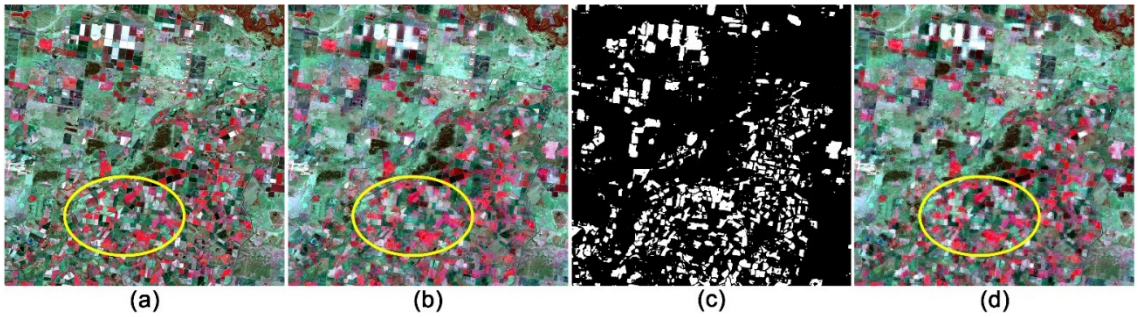


Fig. 8. (a) Actual Landsat images captured on April 26, 2002 in the CIA study site; (b) Unmixed Landsat-like image in the first unmixing process; (c) Land cover change mask detected at the Landsat resolution (white patches: changed and black patches: unchanged); (d) Unmixed Landsat-like image in the second unmixing process. All color images are presented as Landsat bands 4, 1, and 2 displayed as RGB.

of complex land cover changes than the other five models. However, the proposed algorithm is inherently a pixel-based approach, and the basic unmixing process always relies on the input land surface reflectance of low-spatial-resolution images. In contrast, existing learning-based STIF models, as a state-of-the-art technology, conduct the image fusion at a feature level, which has a stronger tolerance to imperfections in the image registration accuracy [29]. To date, these learning-based models have not been popular due to the complexity of their computations [42], [43]. Therefore, the appropriate integration and development of pixel-level, feature-level, and even decision-level models should become a trend in the future research to further improve the accuracy and effectiveness of previous STIF models.

VI. Conclusion

It has always been challenging for existing STIF models to recover the detailed spatial information for complex land cover changes. In this study, LWU-STFM is proposed to specifically address land cover changes using only one pair of prior high- and low-spatial-resolution images. Here, we first define the class fractions at t_p by recomposing the land use classification map. Additionally, we locally conduct the first and second spatial unmixing processes using the developed GSWR module, followed by the idea of neighboring prediction using the optimized similar pixels. All these improvements contribute to better handling complex land cover changes in both homogeneous and heterogeneous landscapes.

LWU-STFM is compared with classic STARFM, FSDAF, SFSDAF, FSDAF 2.0, and VIPSTF-SW to demonstrate the performance for both the semisimulated Landsat–MODIS-like image pair and actual time-series satellite datasets. The quantitative results illustrate that LWU-STFM outperforms the other five models with the highest CC and SSIM and the lowest RMSE and ERGAS. From the visual comparisons, LWU-STFM shows encouraging improvements in effectively capturing complex land cover change areas and then effectively recovering the spatial details. Therefore, the blended images generated by LWU-STFM could provide an alternative to actual satellite imagery with high spatial resolution at a certain date in practical remote sensing applications.

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